

Mathematics of Machine Learning

Singular Value Decomposition (SVD)

The diagram illustrates the Singular Value Decomposition (SVD) of a matrix X . On the left, a large matrix is shown in block form, enclosed in large parentheses. It consists of a top-left block labeled 0 , a top-right block labeled X (highlighted in pink), a bottom-left block labeled X^T (highlighted in blue), and a bottom-right block labeled 0 . This matrix is equated to the product of three matrices. The first matrix is a tall, narrow matrix enclosed in parentheses, with a top section labeled u (highlighted in blue) and a bottom section labeled v (highlighted in pink). This is followed by an equals sign and a scalar s . The final matrix is another tall, narrow matrix enclosed in parentheses, with a top section labeled u (highlighted in blue) and a bottom section labeled v (highlighted in pink).

Singular Valued Decomposition, SVD, general linear maps

Outline

1. **Singular Value Decomposition**
2. General Linear Mappings
3. Linear Regression Revisited

$$\begin{pmatrix} 0 & X \\ X^T & 0 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = s \begin{pmatrix} u \\ v \end{pmatrix}$$

Singular Value Decomposition

- One of the most important results in linear algebra is that any matrix $\mathbf{X}$ can be decomposed as

$$\mathbf{X} = \mathbf{U}\mathbf{S}\mathbf{V}^T$$

- If $\mathbf{X}$ is an $m \times p$ matrix then $\mathbf{U}$ is an $m \times m$ orthogonal matrix and $\mathbf{V}$ is a $p \times p$ orthogonal matrix
- And $\mathbf{S}$ is an $m \times p$ matrix with non-zero diagonal elements $S_{ii} = s_i$ known as the **singular values**

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 $\mathbf{U}\mathbf{U}^T = \mathbf{U}^T\mathbf{U} = \mathbf{I}, \mathbf{V}\mathbf{V}^T = \mathbf{V}^T\mathbf{V} = \mathbf{I}$
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- And $\mathbf{S}$ is an $m \times p$ matrix with non-zero diagonal elements $S_{ii} = s_i$ known as the **singular values**
- Where did that come from?

Eigen Decomposition

- Consider an arbitrary $m \times p$ matrix $\mathbf{X}$, and construct the $(m + p) \times (m + p)$ symmetric matrix, $\mathbf{B}$,

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- Note that as $\mathbf{X}\mathbf{v} = s\mathbf{u}$ and $\mathbf{X}^\top\mathbf{u} = s\mathbf{v}$ then

$$\mathbf{X}(-\mathbf{v}) = (-s)\mathbf{u} \qquad \mathbf{X}^\top\mathbf{u} = (-s)(-\mathbf{v})$$

if $\begin{pmatrix} \mathbf{u} \\ \mathbf{v} \end{pmatrix}$ is an eigenvector of $\mathbf{B}$ with eigenvalue s then so is $\begin{pmatrix} \mathbf{u} \\ -\mathbf{v} \end{pmatrix}$ with eigenvalue $-s$

- If $m < p$ then $\mathbf{X}^\top\mathbf{X}$ is not full rank so some eigenvalues are zero
- As a consequence $p - m$ vectors exist such that $\mathbf{X}\mathbf{v} = 0$
- The eigenvalues and eigenvectors are

$$m \times \left(s_i, \begin{pmatrix} \mathbf{u}_i \\ \mathbf{v}_i \end{pmatrix} \right) \quad m \times \left(-s_i, \begin{pmatrix} \mathbf{u}_i \\ -\mathbf{v}_i \end{pmatrix} \right) \quad (p - m) \times \left(0, \begin{pmatrix} 0 \\ \mathbf{v}_k \end{pmatrix} \right)$$

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Matrix Decomposition

- Stacking the eigenvectors into a matrix

$$\begin{pmatrix} 0 & x \\ x^T & 0 \end{pmatrix} = \begin{pmatrix} u & u & 0 \\ v & -v & v_0 \end{pmatrix} \begin{pmatrix} u & u & 0 \\ v & -v & v_0 \end{pmatrix} \begin{pmatrix} s & 0 & 0 \\ 0 & -s & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

- Since the vectors $\begin{pmatrix} u_i \\ v_i \end{pmatrix}$ are eigenvectors of a symmetric matrix they form an orthogonal matrix if they are normalised.
- Multiply on the right by the transpose of the orthogonal matrix

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Normalisation Subtlety

$$\begin{pmatrix} 0 & \mathbf{X} \\ \mathbf{X}^\top & 0 \end{pmatrix} = \begin{pmatrix} \mathbf{U} & \mathbf{U} & 0 \\ \mathbf{V} & -\mathbf{V} & \mathbf{V}_0 \end{pmatrix} \begin{pmatrix} \mathbf{S} & 0 & 0 \\ 0 & -\mathbf{S} & 0 \\ 0 & 0 & 0 \end{pmatrix} \begin{pmatrix} \mathbf{U}^\top & \mathbf{V}^\top \\ \mathbf{U}^\top & -\mathbf{V}^\top \\ 0 & \mathbf{V}_0^\top \end{pmatrix}$$

- Multiplying out we have

$$\mathbf{X} = 2\mathbf{U}\mathbf{S}\mathbf{V}^\top$$

$$\mathbf{X}^\top = 2\mathbf{V}\mathbf{S}^\top\mathbf{U}^\top$$

- Now the vectors $\mathbf{u}_i$ and $\mathbf{v}_i$ form orthogonal sets as they satisfy

$$\mathbf{X}^\top\mathbf{X}\mathbf{v} = s^2\mathbf{v}$$

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- But they are not normalised (since $\begin{pmatrix} \mathbf{u}_i \\ \mathbf{v}_i \end{pmatrix}$ is normalised). If we define $\tilde{\mathbf{U}} = \sqrt{2}\mathbf{U}$ and $\tilde{\mathbf{V}} = \sqrt{2}\mathbf{V}$ we find

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 - ★ $\mathbf{U}$, $\mathbf{V}$ are orthogonal matrices
 - ★ $\mathbf{S} = \text{diag}(s_1, s_2, \dots, s_r)$ with $r = \min(m, p)$
- s_i can always be chosen to be non-negative and are known as **singular values**
- If $\mathbf{X}$ is symmetric and positive semi-definite (e.g. a covariance matrix) then $\mathbf{U} = \mathbf{V}$ and $\mathbf{S} = \mathbf{\Lambda}$ (the matrix of eigenvalues)
- Singular value decomposition applies to both square and non-square matrices—they describe general linear mappings

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Spectrum of a Matrix

- The set of singular values (when ordered) is often referred to as the (singular value) spectrum of a matrix
- The largest singular value, s_{max} , provides a measure of the maximum change in the two norm of a vector

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- $s_{max} = \|\mathbf{X}\|_2$ is known as the **spectral norm** or **2 norm** of a matrix satisfying

$$\|\mathbf{X}\mathbf{x}\|_2 \leq s_{max}\|\mathbf{x}\|_2$$

Spectrum of a Matrix

- The set of singular values (when ordered) is often referred to as the (singular value) spectrum of a matrix
- The largest singular value, s_{max} , provides a measure of the maximum change in the two norm of a vector

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with equality when $\mathbf{x} = \mathbf{v}_{max}$

Finding SVD

- Most libraries will compute the SVD for you
- In principle we can choose the smaller of the two matrices $\mathbf{X}\mathbf{X}^T$ and $\mathbf{X}^T\mathbf{X}$ and compute its eigenvalues (libraries avoid this: it squares the condition number)
- The singular values are the square root of the eigenvalues (notice that $\mathbf{X}\mathbf{X}^T$ and $\mathbf{X}^T\mathbf{X}$ are both positive semi-definite so the eigenvalues will be non-negative)
- We can then compute the $\mathbf{U}$ matrix or $\mathbf{V}$ matrix by multiplying through by $\mathbf{X}$ or $\mathbf{X}^T$ ($\mathbf{U} = \mathbf{X}\mathbf{V}\mathbf{S}^{-1}$ and $\mathbf{V} = \mathbf{X}^T\mathbf{U}\mathbf{S}^{-1}$)
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Economical Forms of SVD

- Often the rows or columns of the orthogonal matrices $\mathbf{U}$ and $\mathbf{V}$ that are not associated with a singular value are ignored

$$\begin{matrix} \mathbf{X} & = & \mathbf{U} & \mathbf{S} & \mathbf{V}^T \\ \left(\begin{array}{c} \text{ } \\ \text{ } \\ \text{ } \end{array} \right) & = & \left(\begin{array}{c} \text{ } \\ \text{ } \\ \text{ } \end{array} \right) & \left(\begin{array}{cc} \text{ } & 0 \\ 0 & \text{ } \end{array} \right) & \left(\begin{array}{c} \text{ } \\ \text{ } \\ \text{ } \end{array} \right) \end{matrix}$$

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- In numpy (which returns a vector $\mathbf{s}$ and $\mathbf{V}^T$)

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U, s, Vt = np.linalg.svd(X)      # full: U is m x m
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Outline

1. Singular Value Decomposition
2. **General Linear Mappings**
3. Linear Regression Revisited

$$\begin{pmatrix} 0 & X \\ X^T & 0 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = s \begin{pmatrix} u \\ v \end{pmatrix}$$

General Matrix

- Recall that we can compute the SVD for any matrix, $\mathbf{X}$
- As matrices describe the most general linear mapping

$$\mathbf{v} \rightarrow \mathcal{T}[\mathbf{v}] = \mathbf{X}\mathbf{v}$$

- We can use SVD to understand any linear mapping
- Thus any linear mapping can be seen as a rotation followed by a squashing or expansion independently in each coordinate followed by another rotation

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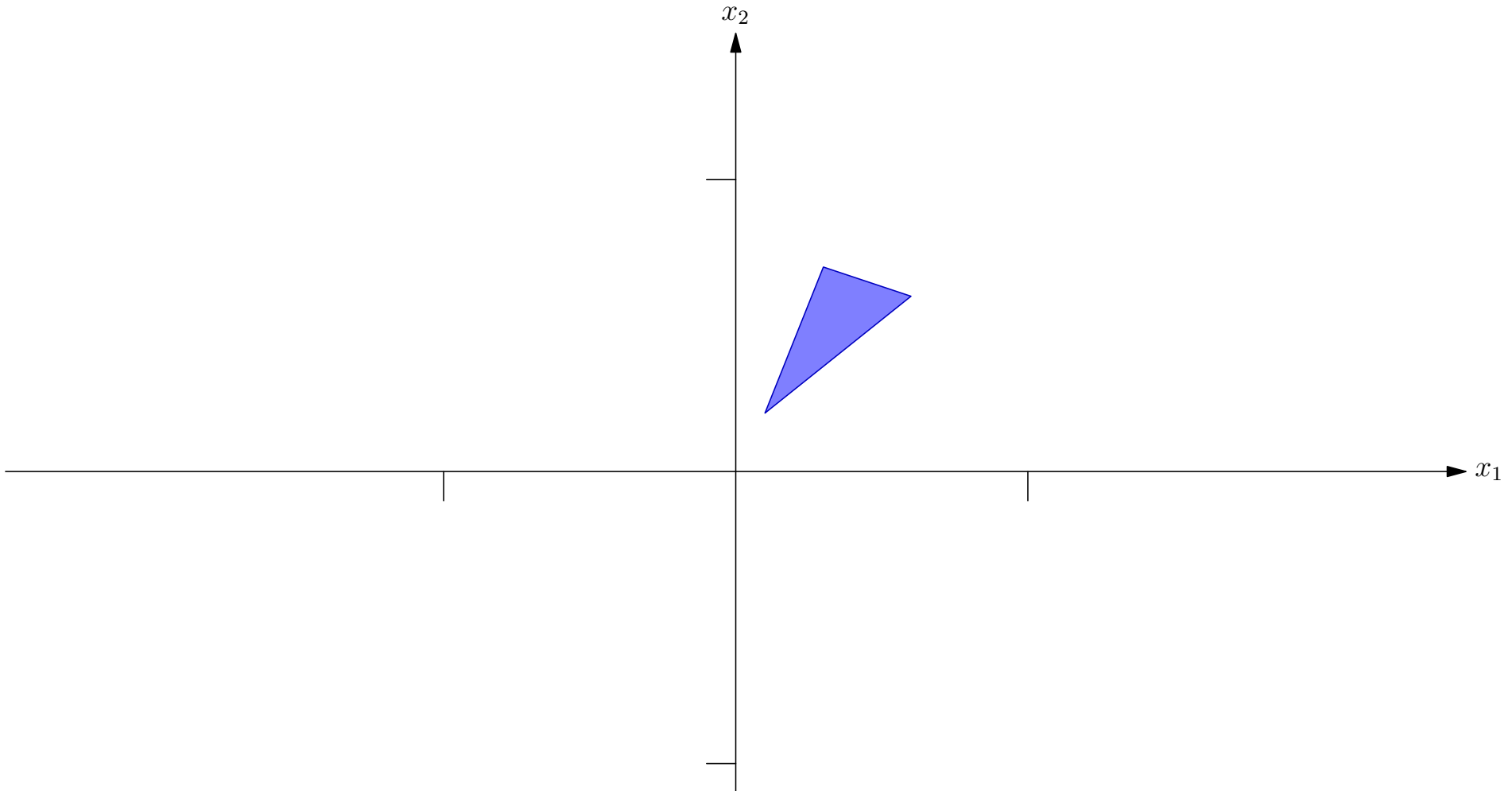
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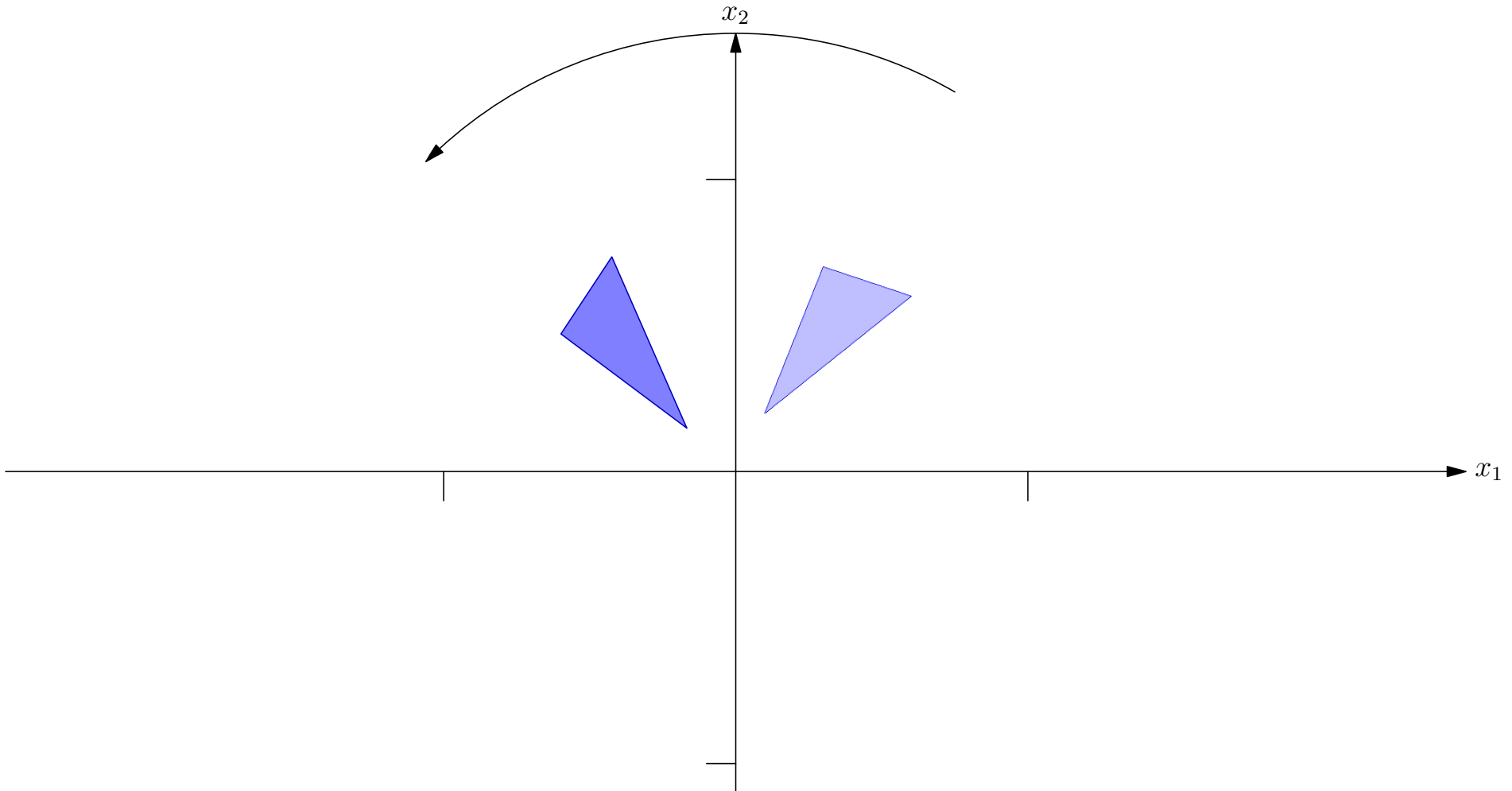
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$$\mathbf{M} = \begin{pmatrix} -0.45 & 1.9 \\ -0.77 & -0.025 \end{pmatrix} = \mathbf{U} \mathbf{S} \mathbf{V}^T = \begin{pmatrix} \cos(-175^\circ) & -\sin(-175^\circ) \\ \sin(-175^\circ) & \cos(-175^\circ) \end{pmatrix} \begin{pmatrix} 2 & 0 \\ 0 & 0.75 \end{pmatrix} \begin{pmatrix} \cos(75^\circ) & -\sin(75^\circ) \\ \sin(75^\circ) & \cos(75^\circ) \end{pmatrix}$$



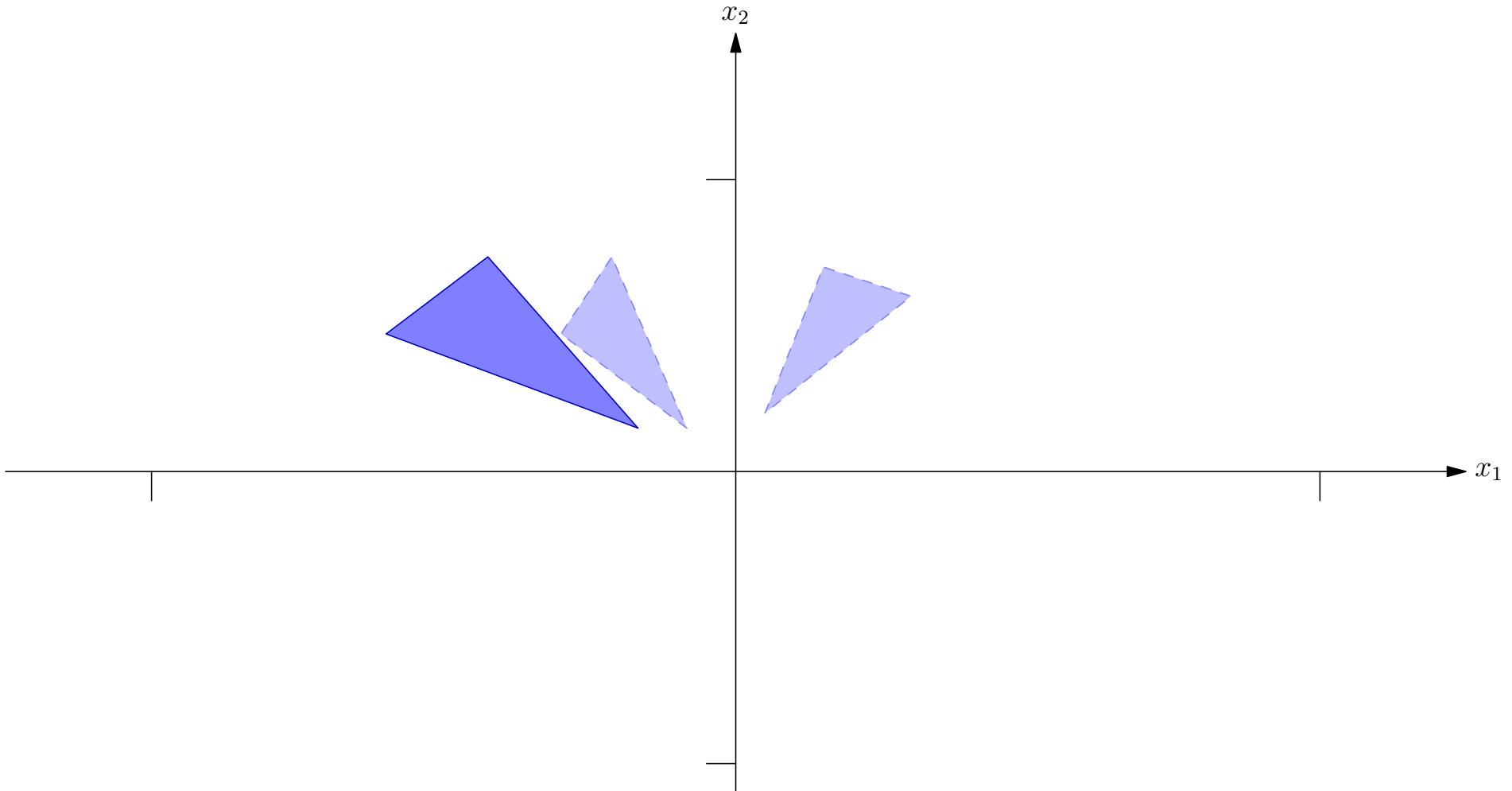
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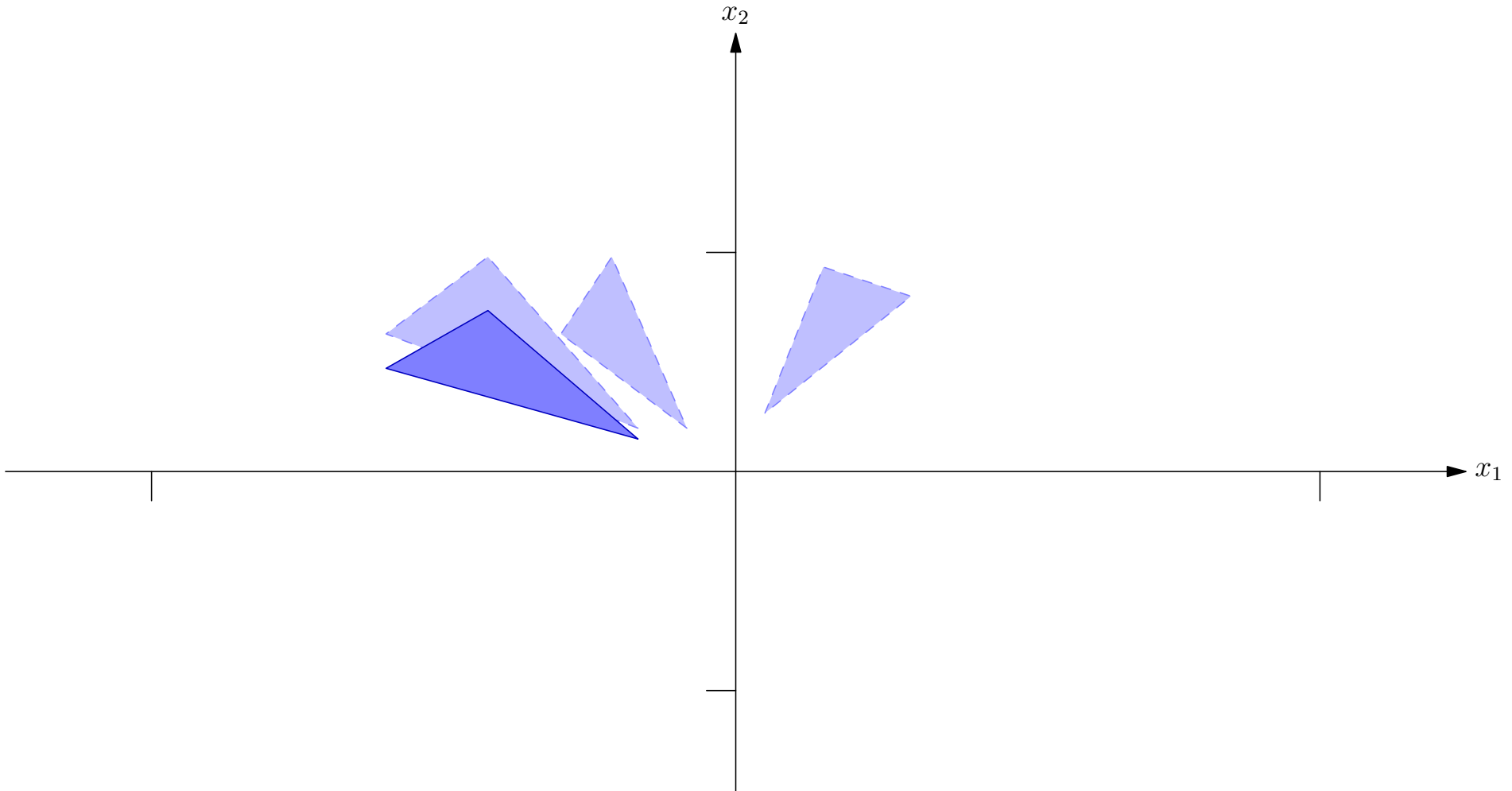
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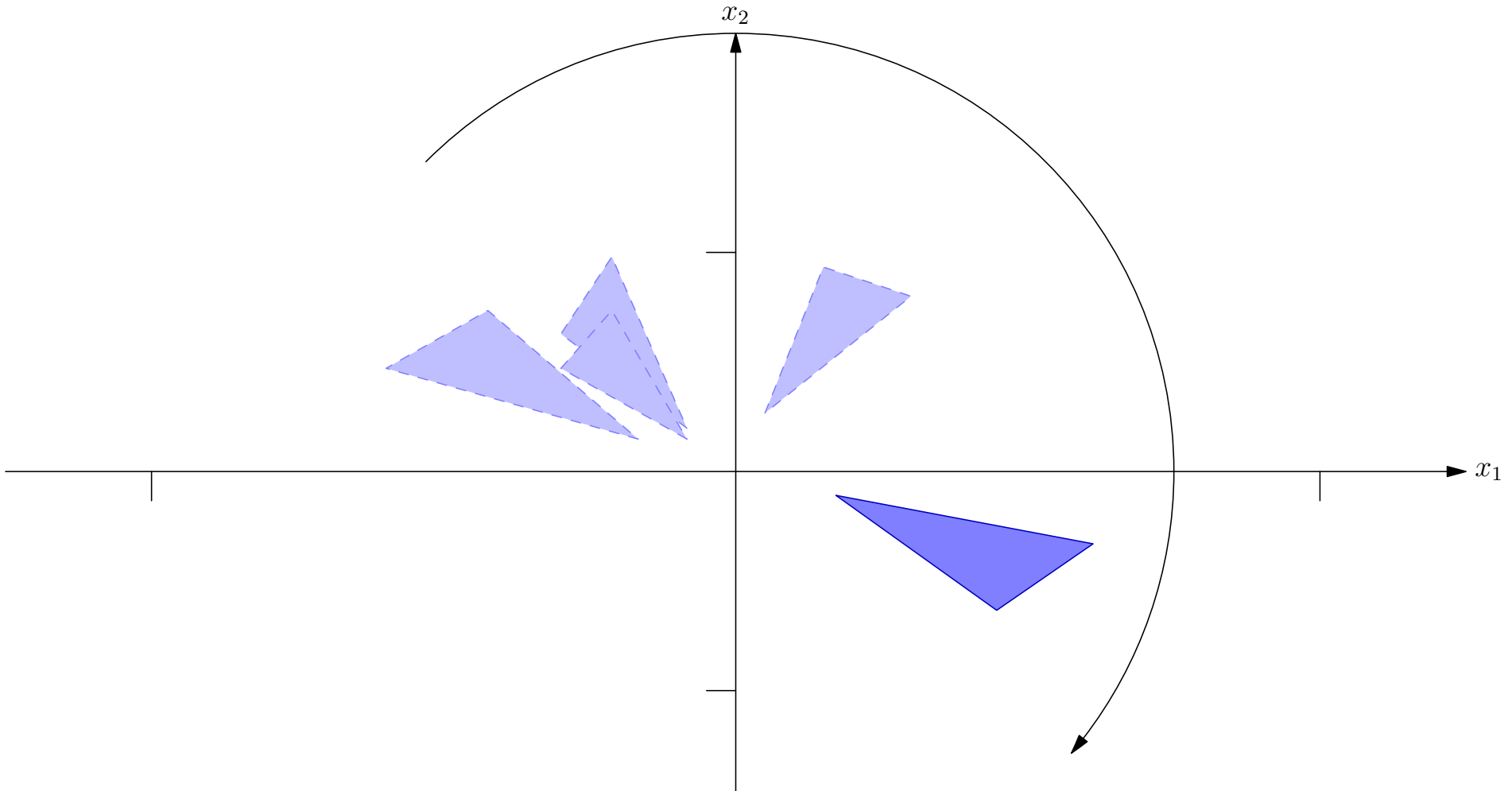
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- The determinant, $|\mathbf{M}|$ of a matrix $\mathbf{M}$ is defined for square matrices
- It describes the change in volume under the mapping
- Now for any two matrices $|\mathbf{AB}| = |\mathbf{A}||\mathbf{B}|$
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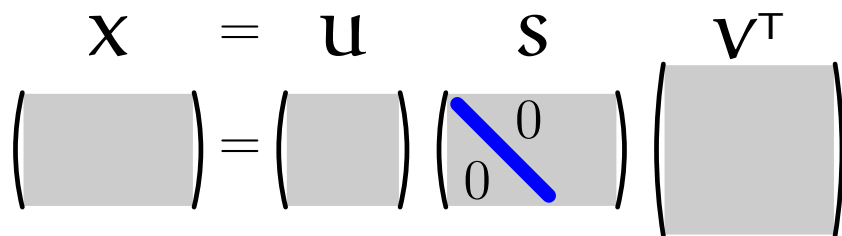
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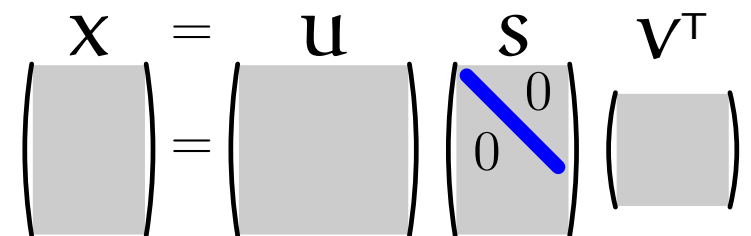
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- When the matrices are non-square then the singular value matrix will either
 - ★ Squash some directions to zero
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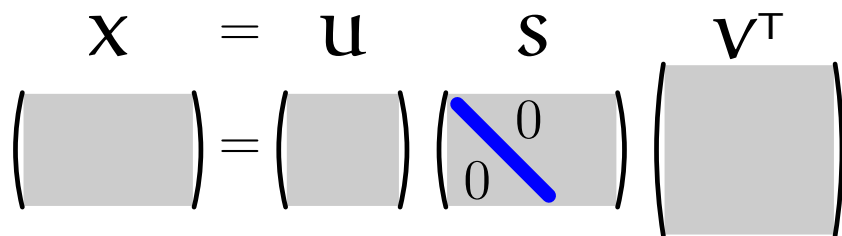
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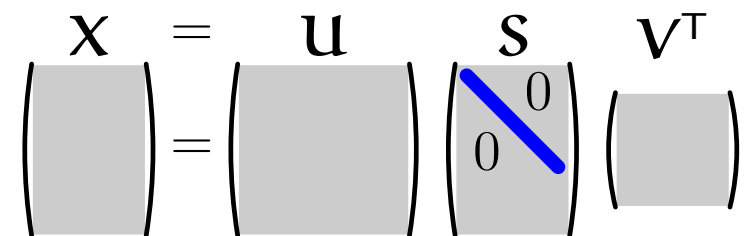
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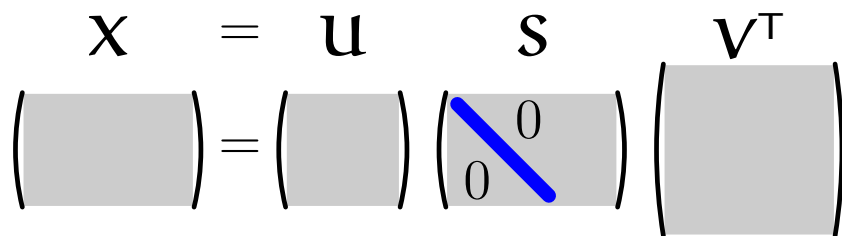
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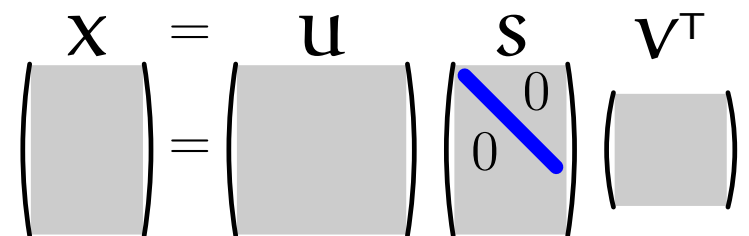
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- $\mathbf{S}^T\mathbf{S}$ is a $p \times p$ matrix with elements $S_{ii}^2 = s_i^2$
- $\mathbf{U}$ and $\mathbf{V}$ are matrices of eigenvectors for $\mathbf{C}$ and $\mathbf{D}$
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Duality Revisited

- If $\mathbf{X} = \mathbf{U}\mathbf{S}\mathbf{V}^T$ then

$$\begin{aligned}\mathbf{C} &= \mathbf{X}\mathbf{X}^T \\ &= \mathbf{U}\mathbf{S}\mathbf{V}^T\mathbf{V}\mathbf{S}^T\mathbf{U}^T \\ &= \mathbf{U}(\mathbf{S}\mathbf{S}^T)\mathbf{U}^T\end{aligned}$$

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SS^T and $S^T S$

$$\mathbf{S} = \begin{pmatrix} s_1 & 0 & \cdots & 0 & 0 & \cdots & 0 \\ 0 & s_2 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & s_m & 0 & 0 \cdots & 0 \end{pmatrix}$$

$$\mathbf{S}^T \mathbf{S} = \begin{pmatrix} s_1^2 & 0 & \cdots & 0 & 0 & \cdots & 0 \\ 0 & s_2^2 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & s_m^2 & 0 & 0 \cdots & 0 \\ 0 & 0 & \cdots & 0 & 0 & 0 \cdots & 0 \\ \vdots & \vdots & \cdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & 0 & 0 \cdots & 0 \end{pmatrix}$$

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$$S^T S = \begin{pmatrix} s_1^2 & 0 & \cdots & 0 & 0 & \cdots & 0 \\ 0 & s_2^2 & \cdots & 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & s_m^2 & 0 & 0 \cdots & 0 \\ 0 & 0 & \cdots & 0 & 0 & 0 \cdots & 0 \\ \vdots & \vdots & \cdots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 & 0 & 0 \cdots & 0 \end{pmatrix}$$

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SS^T and S^TS

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Having A Go

- It's really easy to verify this in Python

```
import numpy as np
X = np.random.rand(3, 2)
U, s, Vt = np.linalg.svd(X)      # s: vector, Vt: V transposed
S = np.zeros_like(X)             # rebuild the 3 x 2 matrix S
np.fill_diagonal(S, s)
U @ S @ Vt                       # should equal X
U[:, 0] @ U[:, 1]                # columns of U are orthogonal
U.T @ U, U @ U.T                # both the identity
L, Ua = np.linalg.eigh(X @ X.T)  # eigenvalues, ascending
S @ S.T                          # the same values: s squared
```

- Test yourself!

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Outline

1. Singular Value Decomposition
2. General Linear Mappings
3. **Linear Regression Revisited**

$$\begin{pmatrix} 0 & X \\ X^T & 0 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = s \begin{pmatrix} u \\ v \end{pmatrix}$$

Linear Regression

- Given a set of data $\mathcal{D} = \{(\mathbf{x}_k, y_k) | k = 1, 2, \dots, m\}$
- In linear regression we try to fit a linear model

$$f(\mathbf{x}|\mathbf{w}) = \mathbf{x}^\top \mathbf{w}$$

- Which we fit by minimising the squared error loss

$$L(\mathbf{w}) = \sum_{k=1}^m (f(\mathbf{x}_k|\mathbf{w}) - y_k)^2$$

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Matrix Form

- In matrix form we write $L(\mathbf{w}) = \|\mathbf{X}\mathbf{w} - \mathbf{y}\|^2$

$$\mathbf{X} = \begin{pmatrix} \mathbf{x}_1^\top \\ \mathbf{x}_2^\top \\ \vdots \\ \mathbf{x}_m^\top \end{pmatrix} \quad \mathbf{y} = \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_m \end{pmatrix}$$

- Then $\nabla L(\mathbf{w}^*) = 0$ implies

$$\mathbf{w}^* = (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{y} = \mathbf{X}^+ \mathbf{y}$$

- This is known as the pseudo-inverse

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- Using $\mathbf{X} = \mathbf{U}\mathbf{S}\mathbf{V}^T$ then

$$\begin{aligned}\mathbf{X}^+ &= (\mathbf{X}^T \mathbf{X})^{-1} \mathbf{X}^T \\ &= (\mathbf{V} \mathbf{S}^T \mathbf{S} \mathbf{V}^T)^{-1} \mathbf{V} \mathbf{S}^T \mathbf{U}^T \\ &= \mathbf{V} (\mathbf{S}^T \mathbf{S})^{-1} \mathbf{V}^T \mathbf{V} \mathbf{S}^T \mathbf{U}^T \\ &= \mathbf{V} (\mathbf{S}^T \mathbf{S})^{-1} \mathbf{S}^T \mathbf{U}^T = \mathbf{V} \mathbf{S}^+ \mathbf{U}^T\end{aligned}$$

- If $m > p$

$$\mathbf{X}^T = \left(\begin{array}{c|c|c|c|c|c|c|c} \text{bar} & \text{bar} & \text{bar} & \text{bar} & \text{bar} & \text{bar} & \text{bar} & \text{bar} \end{array} \right), \mathbf{S}^T = \begin{pmatrix} s_1 & 0 & 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & s_2 & 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & 0 & s_3 & \dots & 0 & & 0 \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & s_p & 0 & 0 \dots & 0 \end{pmatrix}$$

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$$\begin{aligned}\mathbf{X}^+ &= (\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top \\ &= (\mathbf{V} \mathbf{S}^\top \mathbf{S} \mathbf{V}^\top)^{-1} \mathbf{V} \mathbf{S}^\top \mathbf{U}^\top \\ &= \mathbf{V} (\mathbf{S}^\top \mathbf{S})^{-1} \mathbf{V}^\top \mathbf{V} \mathbf{S}^\top \mathbf{U}^\top \\ &= \mathbf{V} (\mathbf{S}^\top \mathbf{S})^{-1} \mathbf{S}^\top \mathbf{U}^\top = \mathbf{V} \mathbf{S}^+ \mathbf{U}^\top\end{aligned}$$

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$$\mathbf{X}^T = \left(\begin{array}{c|c|c|c|c|c|c|c} \text{gray bar} & \text{gray bar} & \text{gray bar} & \text{gray bar} & \text{gray bar} & \text{gray bar} & \text{gray bar} & \text{gray bar} \end{array} \right), \mathbf{S}^T = \begin{pmatrix} s_1 & 0 & 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & s_2 & 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & 0 & s_3 & \dots & 0 & & 0 \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & s_p & 0 & 0 \dots & 0 \end{pmatrix}$$

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Pseudo-Inverse of $\mathbf{S}$

$$\mathbf{S}^\top \mathbf{S} = \begin{pmatrix} s_1^2 & 0 & \dots & 0 \\ 0 & s_2^2 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & s_p^2 \end{pmatrix} \quad (\mathbf{S}^\top \mathbf{S})^{-1} = \begin{pmatrix} s_1^{-2} & 0 & \dots & 0 \\ 0 & s_2^{-2} & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & s_p^{-2} \end{pmatrix}$$

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Moore–Penrose Pseudo-Inverse

- $(\mathbf{X}^\top \mathbf{X})^{-1} \mathbf{X}^\top$ needs every $s_i > 0$: it fails if $m < p$ or the columns of $\mathbf{X}$ are dependent
- The **Moore–Penrose** pseudo-inverse, $\mathbf{X}^+ = \mathbf{V} \mathbf{S}^+ \mathbf{U}^\top$, just skips the zero singular values: $S_{ii}^+ = 1/s_i$ if $s_i \neq 0$, and 0 otherwise
- **Over-constrained** ($m > p$, all $s_i > 0$): nothing changes, $\mathbf{w} = \mathbf{X}^+ \mathbf{y}$ minimises $\|\mathbf{X} \mathbf{w} - \mathbf{y}\|^2$
- **Under-constrained** ($m < p$): infinitely many $\mathbf{w}$ fit the data, and $\mathbf{X}^+ \mathbf{y}$ picks the one with the smallest $\|\mathbf{w}\|$
 - ★ If $s_i = 0$ then $\mathbf{X} \mathbf{v}_i = \mathbf{0}$: the data say nothing about $\mathbf{w}$ along $\mathbf{v}_i$, so that component is set to zero
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III-Conditioned Data Matrix

- Recall that

$$\mathbf{w}^* = \mathbf{X}^+ \mathbf{y} = \mathbf{V} \mathbf{S}^+ \mathbf{U}^T \mathbf{y}$$

- If any of the singular values of $\mathbf{X}$ are small then $\mathbf{S}^+$ will magnify components in that direction
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- This leads to poor weights

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Regularisation

- Consider linear regression with a regulariser

$$\begin{aligned}\mathcal{L}(\mathbf{w}) &= \|\mathbf{X}\mathbf{w} - \mathbf{y}\|^2 + \eta\|\mathbf{w}\|^2 \\ &= \mathbf{w}^\top (\mathbf{X}^\top \mathbf{X} + \eta \mathbf{I}) \mathbf{w} - 2\mathbf{w}^\top \mathbf{X}^\top \mathbf{y} + \mathbf{y}^\top \mathbf{y}\end{aligned}$$

- Thus

$$\nabla \mathcal{L}(\mathbf{w}) = 2(\mathbf{X}^\top \mathbf{X} + \eta \mathbf{I}) \mathbf{w} - 2\mathbf{X}^\top \mathbf{y}$$

- and $\nabla \mathcal{L}(\mathbf{w}^*) = 0$ gives

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Regularisation Continued

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$$(\mathbf{S}^\top \mathbf{S} + \eta \mathbf{I})^{-1} \mathbf{S}^\top = \begin{pmatrix} \frac{s_1}{s_1^2 + \eta} & 0 & 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & \frac{s_2}{s_2^2 + \eta} & 0 & \dots & 0 & 0 & \dots & 0 \\ 0 & 0 & \frac{s_3}{s_3^2 + \eta} & \dots & 0 & & 0 \dots & 0 \\ \vdots & \vdots & \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & 0 & \dots & \frac{s_p}{s_p^2 + \eta} & 0 & 0 \dots & 0 \end{pmatrix}$$

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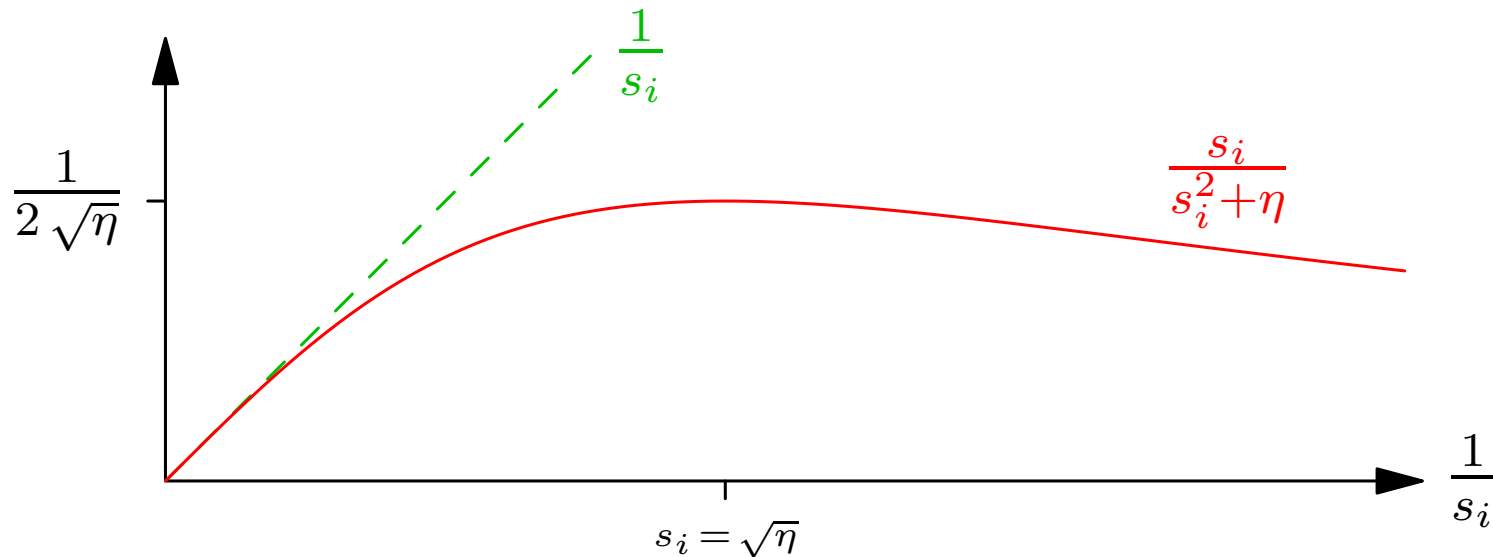
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Effect of Regularisation

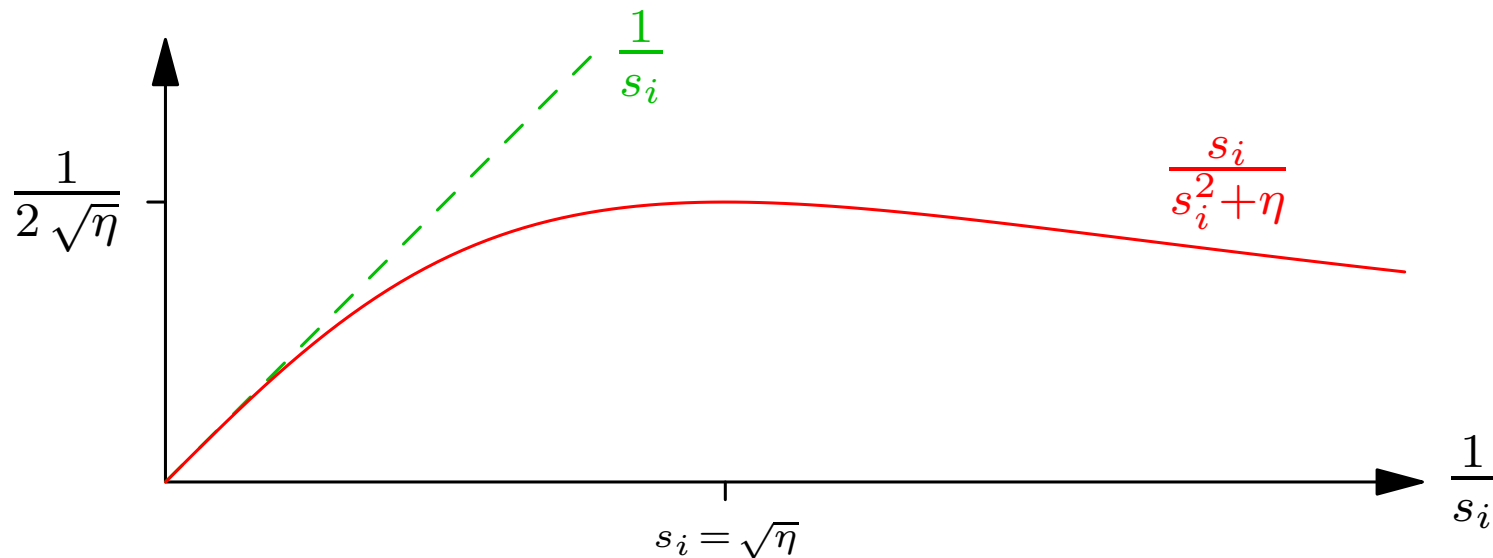
- Without regularisation if $s_i = 0$ the problem would be ill-posed (even $\mathbf{S}^+$ does not exist since s_i^{-1} would be ill defined) and if s_i is small then $\mathbf{S}^+$ is ill-conditioned
- Using $\hat{\mathbf{S}}^+ = (\mathbf{S}^\top \mathbf{S} + \eta \mathbf{I})^{-1} \mathbf{S}^\top$ instead of $\mathbf{S}^+$ then



- Regularisation makes the machine much more stable (reduces the variance)

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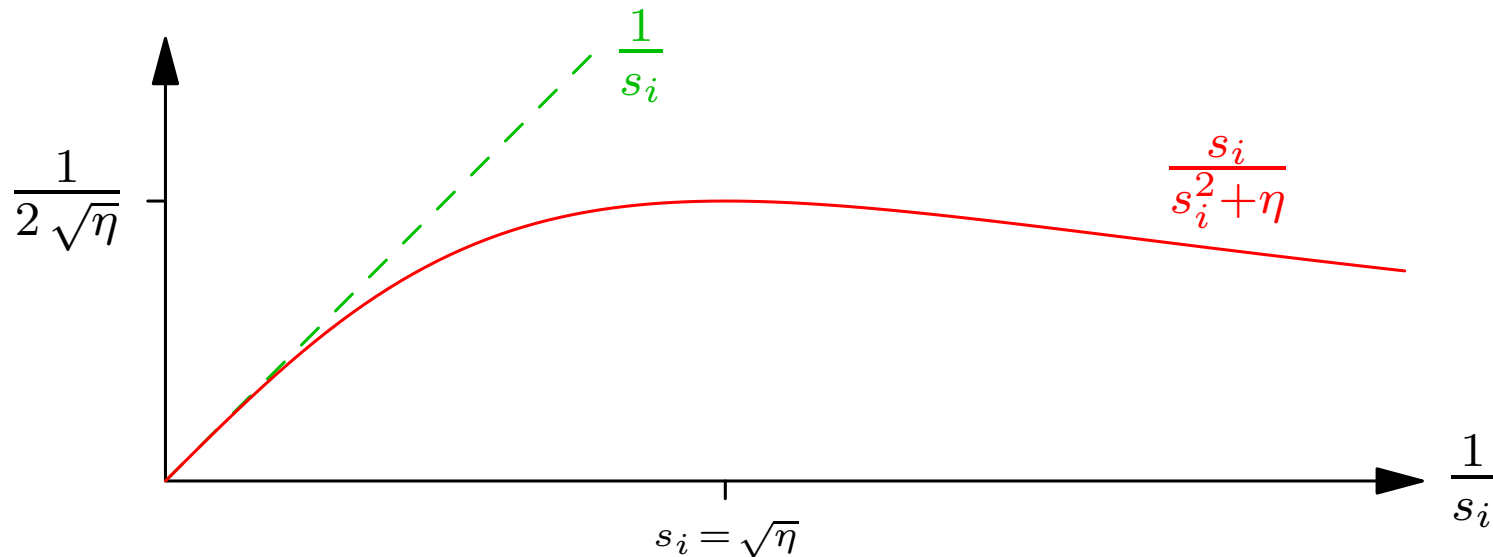
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Summary

- Any matrix can be decomposed as $\mathbf{X} = \mathbf{U}\mathbf{S}\mathbf{V}^T$ where
 - ★ $\mathbf{U}$ and $\mathbf{V}$ are orthogonal (rotation matrices)
 - ★ $\mathbf{S} = \text{diag}(s_1, \dots, s_r)$ is a diagonal matrix of non-negative singular values
- This describes the most general linear transform
- The transform exploits the duality between $\mathbf{X}\mathbf{X}^T$ and $\mathbf{X}^T\mathbf{X}$
- In linear regression the pseudo-inverse involves the reciprocal of the singular values, which can lead to poor generalisation
- Regularisation improves the conditioning of the “inverse” matrix

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